Chapter 10. Inner Product Spaces

Orthonormal bases, orthogonal and unitary groups, adjoints

Contents

10.1 Inner product spaces **2**

10.2 Properties of inner product spaces **4**

10.3 Gram–Schmidt orthogonalization **7**

10.4 Standard basis versus orthonormal basis **9**

10.5 Isomorphisms of inner product spaces **11**

10.6 Orthogonal groups and unitary groups **12**

10.7 The adjoint matrix and its applications **15**

We now leave Euclidean space for general inner product spaces, real or complex, finite- or infinite-dimensional, carrying along the philosophy and the tools of Chapter 9.

10.1 Inner product spaces

Motivation. Extending the dot product to $\mathbb{C}^n$ requires a complex conjugate in the second slot. Abstracting the four properties that all the arguments of Chapter 9 actually used gives the notion of an inner product on any real or complex vector space.

Definition 10.1.1 (Hermitian dot product)

For $X = (a_1, \dots, a_n)^t$, $Y = (b_1, \dots, b_n)^t \in \mathbb{C}^n$,

$$\langle X, Y \rangle = X^t \cdot \bar{Y} = \sum_{i=1}^{n} a_i \bar{b}_i$$

is the (**Hermitian**) **dot product** of $\mathbb{C}^n$.

Remark 10.1.2

Without conjugation, $(1+i, 1-i)^t$ would have "length" 0 ($(1+i)^2 + (1-i)^2 = 0$) and $(i, 0)^t$ an imaginary "length".

Exercise 10.1.3

For $X, Y, Z \in \mathbb{C}^n$, $c \in \mathbb{C}$: (a) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$, $\langle cX, Y\rangle = c\langle X, Y\rangle$;
(b) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$, $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$;
(c) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$;
(d) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$.

Solution. (a), (b) from $\overline{b+c} = \bar{b} + \bar{c}$, $\overline{cb} = \bar{c}\bar{b}$.

(c) $\overline{\sum b_i \bar{a}_i} = \sum \bar{b}_i a_i$.

(d) $\langle X, X\rangle = \sum |a_i|^2$. □

Rereading Chapter 9, almost every argument used only Exercises 9.1.2 and 10.1.3; so we start from these properties.

Definition 10.1.4 (Inner product on F^n)

($F = \mathbb{R}$ or $\mathbb{C}$.) A function $(X, Y) \mapsto \langle X, Y\rangle$, $F^n \times F^n \to F$, is an **inner product** on F^n if for all $X, Y, Z \in F^n$, $c \in F$: (1) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$; (2) $\langle cX, Y\rangle = c\langle X, Y\rangle$; (3) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$; (4) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$. (For $F = \mathbb{C}$, also a **Hermitian inner product**.)

Observation 10.1.5

If $\langle \cdot, \cdot \rangle$ is an inner product on F^n: (a) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$;
(b) $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$.

Proof. By (3), $\langle X, Y+Z\rangle = \overline{\langle Y+Z, X\rangle} = \overline{\langle Y, X\rangle} + \overline{\langle Z, X\rangle} = \langle X, Y\rangle + \langle X, Z\rangle$, and $\langle X, cY\rangle = \overline{c\langle Y, X\rangle} = \bar{c}\,\overline{\langle Y, X\rangle} = \bar{c}\langle X, Y\rangle$. ■

Definition 10.1.6 (Inner product space)

Let V be an F-vector space (possibly infinite-dimensional). $\langle\cdot,\cdot\rangle : V \times V \to F$ is an **inner product** if for all $u, v, w \in V$, $c \in F$: (1) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$; (2) $\langle cv, w\rangle = c\langle v, w\rangle$; (3) $\langle v, w\rangle = \overline{\langle w, v\rangle}$; (4) if $v \neq 0$, then $\langle v, v\rangle \in \mathbb{R}$ and $\langle v, v\rangle > 0$ (*positive definiteness*). V with an inner product is an **inner product space**; for $F = \mathbb{C}$, $\langle\cdot,\cdot\rangle$ is also called **Hermitian**.

Remark 10.1.7

(a) Observation 10.1.5 still holds.
(b) For $F = \mathbb{R}$, conjugation does nothing, so a real inner product is a *symmetric* bilinear form.
(c) For $F = \mathbb{C}$, an inner product is linear in the first variable and **conjugate linear** (anti-linear) in the second.
(d) Every F-subspace of an inner product space is an inner product space.

Exercise 10.1.8

$\langle X, Y\rangle = \sum_{j=1}^{n} j\, a_j b_j$ is an inner product on $\mathbb{R}^n$.

Solution. (1)–(3) are clear; $\langle X, X\rangle = \sum j a_j^2 > 0$ for $X \neq 0$. □

Exercise 10.1.9

$\left\langle \binom{a}{b}, \binom{c}{d} \right\rangle = 2ac - bc - ad + 2bd$ is an inner product on $\mathbb{R}^2$.

Solution. It is bilinear and symmetric, and $\langle (a,b),(a,b)\rangle = 2a^2 - 2ab + 2b^2 = a^2 + b^2 + (a-b)^2 > 0$ for $(a,b) \neq 0$. □

So $\mathbb{R}^n$ carries infinitely many inner products.

Exercise 10.1.10

If $\langle\cdot,\cdot\rangle_1, \langle\cdot,\cdot\rangle_2$ are inner products on V, so is $\langle v, w\rangle = \langle v, w\rangle_1 + \langle v, w\rangle_2$.

Solution. (1)–(3) are additive; for $v \neq 0$ both summands are positive reals. □

Exercise 10.1.11

$\langle f(t), g(t)\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product on $\mathbb{R}[t]$.

Solution. Bilinearity and symmetry are clear. If $f \neq 0$, $f^2 \geq 0$ is continuous and positive except at finitely many points, so $\int_0^1 f^2 > 0$. □

For $f = f_1 + i f_2 : \mathbb{R} \to \mathbb{C}$ (real and imaginary parts), $\int_a^b f = \int_a^b f_1 + i \int_a^b f_2$; f is continuous iff f_1, f_2 are.

Exercise 10.1.12

$\langle f(t), g(t)\rangle = \int_0^1 f(t)\overline{g(t)}\,dt$ is an inner product on $\mathbb{C}[t]$.

Solution. As in Exercise 10.1.11, with $\langle f, f\rangle = \int_0^1 |f|^2$ and $\overline{\int f\bar{g}} = \int \bar{f} g$. □

Exercise 10.1.13

$C^0[0,2\pi] = \{f : [0,2\pi] \to \mathbb{C} \text{ continuous}\}$ is a $\mathbb{C}$-vector space, and $\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}\,dx$ is an inner product on it.

Solution. Sums and multiples of continuous functions are continuous. (1)–(3) as before; if $f \neq 0$, $|f|^2$ is continuous, nonnegative and positive on an interval, so $\langle f, f\rangle > 0$. □

Exercise 10.1.14

$\langle A, B\rangle = \operatorname{tr}(A^t \cdot \bar{B})$ is an inner product on $\mathfrak{M}_{n,n}(F)$ (the **trace form**).

Solution. $\operatorname{tr}(A^t\bar{B}) = \sum_{i,j} A_{ji}\bar{B}_{ji}$ is the dot product of A and B viewed as vectors in F^{n^2}, so Exercise 10.1.3 applies. □

10.2 Properties of inner product spaces

Motivation. Norm, Pythagoras, projection, Cauchy–Schwarz and the triangle inequality hold verbatim in any inner product space. Two questions remain: do orthonormal bases exist, and is $V = W \oplus W^\perp$? Also, over $\mathbb{C}$ distance-preserving maps fixing 0 need not be linear.

Throughout, V is an inner product space over F.

Exercise 10.2.1

If $\langle w, v\rangle = 0$ for all $w \in V$, then $v = 0$.

Solution. Take $w = v$ and use positive definiteness. □

Definition 10.2.2 (Norm)

$\|v\| = \sqrt{\langle v, v\rangle}$.

Exercise 10.2.3

(a) $\|v\| \geq 0$, with equality iff $v = 0$.
(b) $\|cv\| = |c|\|v\|$.
(c) If $v \neq 0$, $\frac{1}{\|v\|}v$ is a **unit vector**.

Solution. (b) $\langle cv, cv\rangle = c\bar{c}\langle v, v\rangle = |c|^2\langle v, v\rangle$. The rest as in Exercise 9.1.5. □

Exercise 10.2.4 (Parallelogram law)

$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2)$.

Solution. $\|v \pm w\|^2 = \|v\|^2 \pm (\langle v, w\rangle + \langle w, v\rangle) + \|w\|^2$; add. □

Definition 10.2.5 (Perpendicularity)

(a) $v \perp w$ if $\langle v, w\rangle = 0$ (equivalently $\langle w, v\rangle = 0$).
(b) For $S, T \subseteq V$, $S \perp T$ if $\langle v, w\rangle = 0$ for all $v \in S$, $w \in T$.

Exercise 10.2.6

If $v \perp w$: (a) $\|v + w\|^2 = \|v\|^2 + \|w\|^2$ (Pythagoras);
(b) $\|v + w\| = \|v - w\|$.

Solution. In the expansion of Exercise 10.2.4 the cross terms $\langle v, w\rangle + \langle w, v\rangle$ vanish. □

Exercise 10.2.7

If $w \neq 0$, then $\left(v - \frac{\langle v,w\rangle}{\langle w,w\rangle} w\right) \perp w$ (compare Example 9.4.1).

Solution. $\langle v, w\rangle - \frac{\langle v,w\rangle}{\langle w,w\rangle}\langle w, w\rangle = 0$. □

Observation 10.2.8

(a) (Cauchy–Schwarz) $|\langle v, w\rangle| \leq \|v\|\|w\|$, with equality iff $\{v, w\}$ is linearly dependent.
(b) (Triangle inequality) $\|v + w\| \leq \|v\| + \|w\|$.

Proof. (a) Trivial if $w = 0$. Otherwise let $a = \frac{\langle v,w\rangle}{\langle w,w\rangle}$. By Exercise 10.2.7 and Pythagoras, $\|v\|^2 = \|v - aw\|^2 + \|aw\|^2 \geq |a|^2\|w\|^2$, i.e. $|a|\|w\| \leq \|v\|$, which is the inequality. Equality holds iff $v = aw$.
(b) Write $\operatorname{Re} z$ for the real part; $z + \bar{z} = 2\operatorname{Re} z \leq 2|z|$. Then $\|v+w\|^2 = \|v\|^2 + \langle v, w\rangle + \overline{\langle v, w\rangle} + \|w\|^2 = \|v\|^2 + 2\operatorname{Re}\langle v, w\rangle + \|w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$. ■

So $\|v - w\|$ is a distance on V: every inner product space carries a geometry.

Exercise 10.2.9

(a) $\|v - v\| = 0$.
(b) $\|v - w\| = \|w - v\|$.
(c) $\|v - w\| \leq \|v - u\| + \|u - w\|$.

Solution. As in Exercise 9.1.8, using Observation 10.2.8(b). □

Definition 10.2.10 (Orthogonal and orthonormal sets)

(a) Nonzero vectors $\{v_i\}_{i\in I}$ with $v_i \perp v_j$ for $i \neq j$ form an **orthogonal subset**; if all are unit vectors, an **orthonormal subset**.
(b) For $W \leq V$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, an **orthonormal basis** if it is orthonormal.

Exercise 10.2.11

An orthogonal subset $\{v_i\}_{i\in I}$ is linearly independent.

Solution. If $\sum_{\text{finite}} c_i v_i = 0$, take the inner product with v_j: $c_j\|v_j\|^2 = 0$. □

Exercise 10.2.12

A basis $\{v_i\}$ of $W \leq V$ is orthonormal iff $\langle v_i, v_j\rangle = \delta_{ij}$.

Solution. This restates the definition. □

Exercise 10.2.13

(a) $\{(0,-i)^t, (i,0)^t\}$ is an orthonormal basis of $\mathbb{C}^2$ (dot product).
(b) $\{\frac{1}{\sqrt{3}}(1+i,i)^t, \frac{1}{\sqrt{6}}(1-i,-2)^t\}$ is an orthonormal basis of $\mathbb{C}^2$.
(c) $\{\frac{1}{\sqrt{2}}(1,0)^t, \frac{1}{\sqrt{6}}(1,2)^t\}$ is an orthonormal basis of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9.
(d) $\{1, \sqrt{12}(t-\frac{1}{2})\}$ is an orthonormal basis of $\mathcal{P}_1(\mathbb{R})$ with the inner product of Exercise 10.1.11.
(e) For $f_n(x) = e^{inx} = \cos nx + i \sin nx$ $(n \in \mathbb{Z})$, $\{f_n\}_{n\in\mathbb{Z}}$ is an orthonormal subset of $C^0[0, 2\pi]$ (Exercise 10.1.13).

Solution. (a) Norms 1, and $0 \cdot \bar{i} + (-i) \cdot 0 = 0$. (b) $|1+i|^2 + |i|^2 = 3$, $|1-i|^2 + 4 = 6$, and $(1+i)\overline{(1-i)} + i \cdot \overline{(-2)} = (1+i)^2 - 2i = 0$. (c) $\langle (1,0),(1,0)\rangle = 2$, $\langle (1,2),(1,2)\rangle = 2-2-2+8 = 6$, $\langle (1,0),(1,2)\rangle = 2-0-2+0 = 0$. (d) $\int_0^1 1 = 1$, $\int_0^1 12(t-\frac{1}{2})^2 dt = 1$, $\int_0^1 (t-\frac{1}{2})dt = 0$. (e) $\frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)x}dx = \delta_{mn}$, since for $m \neq n$ the antiderivative $\frac{e^{i(m-n)x}}{i(m-n)}$ is 2π-periodic. □

Question 10.2.14

If V is a (finite-dimensional) inner product space and $W \leq V$, does W always have an orthonormal basis?

Definition 10.2.15 (Orthogonal complement)

$S^\perp = \{v \in V : w \perp v \text{ for all } w \in S\}$; for $W \leq V$, $W^\perp$ is the **orthogonal complement**.

Exercise 10.2.16

For $S \subseteq V$, $W \leq V$ with basis $\mathfrak{C}$: (a) $S^\perp \leq V$;
(b) $S^\perp = \langle S\rangle^\perp$;
(c) $W^\perp = \mathfrak{C}^\perp$.

Solution. As in Observation 9.1.15; in (a) use conjugate linearity: $\langle w, cv\rangle = \bar{c}\langle w, v\rangle = 0$. □

Exercise 10.2.17

For a basis $\mathfrak{B}$ of V and $W \leq V$: (a) $0^\perp = V$, $V^\perp = \mathfrak{B}^\perp = 0$;
(b) $W \cap W^\perp = 0$;
(c) $W \leq (W^\perp)^\perp$.

Solution. As in Exercise 9.1.16, using Exercise 10.2.1. □

Question 10.2.18

If V is finite-dimensional and $W \leq V$, is $\dim W^\perp = \dim V - \dim W$ and $V = W \oplus W^\perp$?

A **rigid motion** of V is $M : V \to V$ with $\|M(v) - M(w)\| = \|v - w\|$. Do Theorem 9.2.11 and Corollary 9.2.12 still hold? For $F = \mathbb{C}$ the answer is no: Observation 9.2.8(c) fails.

Exercise 10.2.19

(a) On $\mathbb{C} = \mathbb{C}^1$ with the dot product, $M(\alpha) = \bar{\alpha}$ is a non-linear rigid motion fixing 0 (M is conjugate linear).
(b) On $\mathbb{C}^2$, $M((\alpha, \beta)^t) = (\bar{\alpha}, \beta)^t$ is a non-linear rigid motion fixing 0, not even conjugate linear.

Solution. (a) $|\bar{\alpha} - \bar{\beta}| = |\overline{\alpha - \beta}| = |\alpha - \beta|$, but $M(i \cdot 1) = -i \neq iM(1)$. (b) $\|M(v) - M(w)\|^2 = |\overline{\alpha - \alpha'}|^2 + |\beta - \beta'|^2 = \|v - w\|^2$; $M(i(1,0)^t) = (-i, 0)^t \neq iM((1,0)^t)$, and $M(i(0,1)^t) = (0, i)^t \neq \bar{i}M((0,1)^t)$. □

Remark 10.2.20

Via the $\mathbb{R}$-isomorphism $\gamma : \mathbb{C} \to \mathbb{R}^2$, $\gamma(a + ib) = (a, b)^t$, the norms (metrics) of $\mathbb{C}$ and $\mathbb{R}^2$ agree: $\|(a, b)^t - (c, d)^t\| = |(a + ib) - (c + id)|$. So rigid motions of $\mathbb{C}$ and of $\mathbb{R}^2$ are the same thing. Conjugation is a reflection of $\mathbb{R}^2$, which is $\mathbb{R}$-linear but not $\mathbb{C}$-linear. Conversely, every $\mathbb{C}$-linear rigid motion of $\mathbb{C}$ (multiplication by $e^{i\theta}$) is a rotation of $\mathbb{R}^2$.

In Chapter 9 we leaned on the standard orthonormal basis of $\mathbb{R}^n$; in general we do not yet know that orthonormal bases exist.

10.3 Gram–Schmidt orthogonalization

Motivation. Subtracting from each vector its components along the previous ones (the projection of Example 9.4.1) turns any independent list into an orthogonal one with the same spans. This answers Questions 9.1.19, 9.4.8, 10.2.14 and 10.2.18 at once, and ties together the dimension theorem, the rank theorem and the "perp theorem".

Theorem 10.3.1 (Gram–Schmidt orthogonalization)

Let V be an inner product space and $\{v_1, \dots, v_r\}$ linearly independent. (a) Put $w_1 = v_1$ and, for $2 \leq i \leq r$,

$$w_i = v_i - \frac{\langle v_i, w_{i-1}\rangle}{\langle w_{i-1}, w_{i-1}\rangle} w_{i-1} - \cdots - \frac{\langle v_i, w_1\rangle}{\langle w_1, w_1\rangle} w_1.$$

Then $\langle v_1, \dots, v_r\rangle = \langle w_1, \dots, w_r\rangle$ and $\{w_1, \dots, w_r\}$ is an orthogonal subset.
(b) Hence $\{\frac{1}{\|w_1\|}w_1, \dots, \frac{1}{\|w_r\|}w_r\}$ is an orthonormal subset.
(c) In particular, if $r = \dim V$, it is an orthonormal basis of V.

Proof. Induction on i: assume $\langle v_1, \dots, v_i\rangle = \langle w_1, \dots, w_i\rangle$ with $\{w_1, \dots, w_i\}$ orthogonal (nonzero vectors). Then $w_{i+1} \in v_{i+1} + \langle w_1, \dots, w_i\rangle$, so $\langle v_1, \dots, v_{i+1}\rangle = \langle w_1, \dots, w_{i+1}\rangle$. Also $w_{i+1} \neq 0$, since otherwise $v_{i+1} \in \langle v_1, \dots, v_i\rangle$. For $j \leq i$, by orthogonality, $\langle w_{i+1}, w_j\rangle = \langle v_{i+1}, w_j\rangle - \frac{\langle v_{i+1}, w_j\rangle}{\langle w_j, w_j\rangle}\langle w_j, w_j\rangle = 0$. (b), (c) follow by normalizing (Exercise 10.2.11). ■

So finite-dimensional inner product spaces and their subspaces have orthonormal bases.

Exercise 10.3.2

Apply Gram–Schmidt to the basis $\mathfrak{B}$: (a) $\mathfrak{B} = \{(1,1,0)^t, (-1,0,2)^t, (2,1,1)^t\}$ of $\mathbb{R}^3$ (dot product); (b) $\mathfrak{B} = \{(3,1,0,1)^t, (0,-1,2,0)^t, (1,1,-1,-1)^t\}$ of a subspace W of $\mathbb{R}^4$; (c) $\mathfrak{B} = \{(1,1)^t, (0,1)^t\}$ of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (d) $\mathfrak{B} = \{1, t, t^2\}$ of $\mathcal{P}_2(\mathbb{R})$ with the inner product of Exercise 10.1.11.

Solution. (a) $w_1 = (1,1,0)$, $w_2 = (-\frac{1}{2},\frac{1}{2},2)$, $w_3 = (\frac{2}{3},-\frac{2}{3},\frac{1}{3})$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1,0)$, $\frac{1}{3\sqrt{2}}(-1,1,4)$, $\frac{1}{3}(2,-2,1)$.

(b) $w_1 = (3,1,0,1)$, $w_2 = \frac{1}{11}(3,-10,22,1)$, $w_3 = \frac{1}{9}(3,2,1,-11)$; orthonormal basis $\frac{1}{\sqrt{11}}(3,1,0,1)$, $\frac{1}{\sqrt{594}}(3,-10,22,1)$, $\frac{1}{\sqrt{135}}(3,2,1,-11)$.

(c) $\langle(1,1),(1,1)\rangle = 2$, $\langle(0,1),(1,1)\rangle = 1$, so $w_2 = (0,1) - \frac{1}{2}(1,1) = (-\frac{1}{2},\frac{1}{2})$ with $\langle w_2, w_2\rangle = \frac{3}{2}$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{6}}(-1,1)$.

(d) 1, $t-\frac{1}{2}$, $t^2-t+\frac{1}{6}$; orthonormal basis 1, $\sqrt{3}(2t-1)$, $\sqrt{5}(6t^2-6t+1)$. □

Exercise 10.3.3

If $\{v_1,\dots,v_r\}$ in Theorem 10.3.1 is already orthogonal, then $w_i = v_i$ for all i.

Solution. All the subtracted terms have $\langle v_i, w_j\rangle = \langle v_i, v_j\rangle = 0$, by induction. □

Corollary 10.3.4 (Perp theorem, inner product case)

If V is a finite-dimensional inner product space and $W \le V$, then $\dim V = \dim W + \dim W^\perp$.

Proof. Extend an orthonormal basis $\{w_1,\dots,w_r\}$ of W to a basis of V and apply Gram–Schmidt; by Exercise 10.3.3 we obtain an orthonormal basis $\{w_1,\dots,w_n\}$ of V beginning with the given one. Then $W^\perp = \langle w_{r+1},\dots,w_n\rangle$: these vectors are perpendicular to $w_1,\dots,w_r$, and if $v = \sum a_i w_i \in W^\perp$, then $a_j = \langle v, w_j\rangle = 0$ for $j \le r$. ■

Corollary 10.3.5

If V is a finite-dimensional inner product space and $W \le V$, then $V = W \oplus W^\perp$.

Proof. $W \cap W^\perp = 0$ (Exercise 10.2.17) and the dimensions add up (Corollary 10.3.4). ■

Exercise 10.3.6

For a finite-dimensional inner product space V and $U, W \le V$: (a) $W = (W^\perp)^\perp$;
(b) $(U+W)^\perp = U^\perp \cap W^\perp$;
(c) $(U \cap W)^\perp = U^\perp + W^\perp$.

Solution. (a) $W \le (W^\perp)^\perp$ and both have dimension $\dim W$ by Corollary 10.3.4 (applied twice). (b) $v \perp U+W$ iff $v \perp U$ and $v \perp W$. (c) Apply (b) to $U^\perp, W^\perp$: $(U^\perp + W^\perp)^\perp = U^{\perp\perp} \cap W^{\perp\perp} = U \cap W$, and take $\perp$ using (a). □

Remark 10.3.7

Cancelling $W^\perp$ from $W \oplus W^\perp = V = W^\perp \oplus (W^\perp)^\perp$ to get (a) is nonsense: complements in a direct sum are not unique (Remark 7.6.10).

With the dot product on $\mathbb{R}^n$, $AX = 0$ reads $\langle [A]_i^t, X\rangle = 0$ for all rows i, so

$$\ker L_A = \langle [A]_1^t,\dots,[A]_m^t\rangle^\perp = [\text{row space of } A]^\perp.$$

Example 10.3.8

For $F = \mathbb{R}$, the perp theorem for the dot product follows from the dimension and rank theorems. For $W = \langle X_1, \dots, X_m \rangle \leq \mathbb{R}^n$, let A have rows X_i^t. Then $W^\perp = \ker L_A$, and by the rank theorem $\dim W = \text{row rank} = \text{column rank} = \dim \operatorname{im} L_A$. The dimension theorem gives $\dim W^\perp = n - \dim W$.

Proposition 10.3.9 (The three pillars, $F = \mathbb{R}$)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on $\mathbb{R}^n$), any two imply the third.

Proof. All three are relations among n, $\dim \ker L_A = \dim W^\perp$ (with W the row space), $\dim W$ (row rank) and $\dim \operatorname{im} L_A$ (column rank): (1) $\dim \ker L_A + \dim \operatorname{im} L_A = n$; (2) $\dim W = \dim \operatorname{im} L_A$; (3) $\dim W + \dim W^\perp = n$. Any two of these linear relations imply the third. ■

In particular this gives a third proof of the rank theorem, one that needs no row reduction. Over $\mathbb{C}$ the "ordinary product" X^tY is not an inner product, and $X \mapsto A\bar{X}$ is not $\mathbb{C}$-linear, so a trick is needed:

Exercise 10.3.10

For $W \leq \mathbb{C}^n$, $\overline{W} = \{\bar{X} : X \in W\}$ is a subspace of $\mathbb{C}^n$ with $\dim \overline{W} = \dim W$.

Solution. $\overline{W}$ is closed under sums and under $c\bar{X} = \overline{\bar{c}X}$. If $\{X_i\}$ is a basis of W, then $\{\bar{X}_i\}$ is a basis of $\overline{W}$ (conjugate any relation). □

Proposition 10.3.11 (The three pillars)

For $F = \mathbb{R}$ or $\mathbb{C}$, of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on F^n), any two imply the third.

Proof. As in Proposition 10.3.9, using $A\bar{X} = 0 \iff \langle [A]_i^t, X \rangle = 0$: the solution space of $A\bar{X} = 0$ is $W^\perp$, and it equals $\overline{\ker L_A}$, of dimension $\dim \ker L_A$ (Exercise 10.3.10). ■

10.4 Standard basis versus orthonormal basis

Motivation. What made the standard basis of $\mathbb{R}^n$ special is that coordinates are inner products and the inner product is the dot product of coordinates. Every orthonormal basis has these properties, which gives orthogonal projections, closest vectors and Bessel's inequality.

For the dot product on F^n and the standard basis $\mathcal{E}$: (a) $X = \sum \langle X, e_i \rangle e_i$;
(b) $\langle X, Y \rangle = [X]_\mathcal{E}^t \overline{[Y]_\mathcal{E}}$.

Observation 10.4.1

If $\mathfrak{B} = \{v_1, \dots, v_n\}$ is an orthogonal basis of V:
(a)′ $v = \sum_i \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} v_i$. If $\mathfrak{B}$ is an orthonormal basis:
(a) $v = \sum_i \langle v, v_i \rangle v_i$, i.e. the i-th coordinate of $[v]_\mathfrak{B}$ is $\langle v, v_i \rangle$;
(b) $\langle v, w \rangle = [v]_\mathfrak{B}^t \cdot \overline{[w]_\mathfrak{B}}$.

Proof. (a) If $v = \sum a_i v_i$, then $\langle v, v_j \rangle = a_j \langle v_j, v_j \rangle$; this gives (a)′ and (a).

(b) For $w = \sum b_i v_i$, $\langle v, w\rangle = \sum_{i,j} a_i \bar{b}_j \langle v_i, v_j\rangle = \sum a_i \bar{b}_i$. ■

So in an orthonormal basis the inner product is "really" the dot product (Example 10.5.4). For example, the foot of the perpendicular from $X = (1,2,3)^t$ to the xy-plane is $(1,2,0)^t = \langle X, e_1\rangle e_1 + \langle X, e_2\rangle e_2$.

Observation 10.4.2

Let W be a finite-dimensional subspace of an inner product space V with orthonormal basis $\{v_1, \ldots, v_m\}$. (a) $V = W \oplus W^\perp$.
(b) (Closest vector problem) For $v \in V$, $w = \langle v, v_1\rangle v_1 + \cdots + \langle v, v_m\rangle v_m \in W$ satisfies $v - w \in W^\perp$, and w is the unique vector of W closest to v.

Proof. (a) With w as in (b), $\langle v - w, v_i\rangle = \langle v, v_i\rangle - \langle v, v_i\rangle = 0$ for all i, so $v - w \in W^\perp$ and $V = W + W^\perp$; also $W \cap W^\perp = 0$.

(b) For $u \in W$, $w - u \in W$ is perpendicular to $v - w$, so by Pythagoras $\|v-u\|^2 = \|v-w\|^2 + \|w-u\|^2 \geq \|v - w\|^2$, with equality iff $u = w$. ■

Exercise 10.4.3

(a) In Euclidean $\mathbb{R}^3$, find an orthonormal basis of $W = \langle (1,1,2)^t\rangle^\perp$ and the vector of W closest to $(1,0,-1)^t$.
(b) Find the foot of the perpendicular from $(1,0,0,0)^t$ to the subspace W of Exercise 10.3.2(b).

Solution. (a) $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{3}}(1,1,-1)$. The closest vector is $X - \frac{\langle X,Y\rangle}{\langle Y,Y\rangle} Y = (1,0,-1) + \frac{1}{6}(1,1,2) = (\frac{7}{6}, \frac{1}{6}, -\frac{2}{3})$ (with $Y = (1,1,2)$).
(b) Summing $\langle v, u_i\rangle u_i$ over the orthonormal basis found there: $(\frac{9}{10}, \frac{4}{15}, \frac{2}{15}, \frac{1}{30})$. □

Exercise 10.4.4

If $\{v_1, \ldots, v_n\}$ is an orthonormal basis and $v = \sum a_i v_i$, then $\|v\|^2 = \sum |a_i|^2 = \sum |\langle v, v_i\rangle|^2$.

Solution. Observation 10.4.1(b) with $w = v$, and (a). □

Observation 10.4.5 (Bessel's inequality)

If $\{v_1, \ldots, v_m\}$ is an orthonormal subset of an inner product space V (possibly infinite-dimensional), then $\sum_{i=1}^m |\langle v, v_i\rangle|^2 \leq \|v\|^2$ for all $v \in V$.

Proof. With $W = \langle v_1, \ldots, v_m\rangle$, Observation 10.4.2 gives $v = w + w'$, $w = \sum \langle v, v_i\rangle v_i \in W$, $w' \in W^\perp$, and by Pythagoras $\|v\|^2 = \|w\|^2 + \|w'\|^2 \geq \|w\|^2 = \sum |\langle v, v_i\rangle|^2$. ■

In infinite dimensions one uses a (large) orthonormal subset $\{v_i\}$ and calls $\langle v, v_i\rangle$ the i-th **Fourier coefficient** of v.

Exercise 10.4.6

With $f_n(x) = e^{inx}$ in $C^0[0, 2\pi]$ (Exercise 10.2.13(e)) and $f(x) = x$: (a) $\langle f, f_0\rangle = \pi$ and $\langle f, f_n\rangle = \frac{i}{n}$ for $n \neq 0$;
(b) $\|f\|^2 = \frac{4}{3}\pi^2$;
(c) Bessel's inequality for $\{f_n\}_{-k \leq n \leq k}$ gives $\sum_{n=1}^k \frac{1}{n^2} \leq \frac{\pi^2}{6}$, hence $\sum_{n=1}^\infty \frac{1}{n^2} \leq \frac{\pi^2}{6}$.

Solution. (a) $\frac{1}{2\pi}\int_0^{2\pi} x\,dx = \pi$; for $n \neq 0$, integrating by parts, $\frac{1}{2\pi}\int_0^{2\pi} xe^{-inx}dx = \frac{1}{2\pi}\left[\frac{xe^{-inx}}{-in}\right]_0^{2\pi} = \frac{1}{-in} = \frac{i}{n}$. (b) $\frac{1}{2\pi}\int_0^{2\pi} x^2 dx = \frac{4\pi^2}{3}$. (c) $\frac{4}{3}\pi^2 \geq \pi^2 + 2\sum_{n=1}^{k}\frac{1}{n^2}$. □

10.5 Isomorphisms of inner product spaces

Motivation. Two inner product spaces are "the same" if a linear isomorphism carries one inner product to the other. Orthonormal bases show that a finite-dimensional inner product space is determined by its dimension: it is F^n with the dot product under another name.

Definition 10.5.1 (Isomorphism of inner product spaces)

Inner product spaces V, W over F are **isomorphic** if there is a vector space isomorphism $\varphi : V \to W$ with $\langle \varphi(u), \varphi(v)\rangle = \langle u, v\rangle$ for all $u, v \in V$; φ is an **inner product space isomorphism**. Equivalently, with $(\varphi \times \varphi)(u, v) = (\varphi(u), \varphi(v))$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{\langle\,,\rangle} & F \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle I} \\ W \times W & \xrightarrow[\langle\,,\rangle]{} & F \end{array}$$

commutes.

Exercise 10.5.2

Is $\varphi \times \varphi : V \times V \to W \times W$ a vector space isomorphism between the product spaces?

Solution. Yes: it is linear componentwise, with inverse $\varphi^{-1} \times \varphi^{-1}$. □

Isomorphic inner product spaces share every property expressible with sums, scalar multiples and inner products.

Exercise 10.5.3

Isomorphism of inner product spaces is an equivalence relation.

Solution. I_V works; inverses of isomorphisms preserve inner products ($\langle \varphi^{-1}u, \varphi^{-1}v\rangle = \langle \varphi\varphi^{-1}u, \varphi\varphi^{-1}v\rangle$); composites of isomorphisms preserve them. □

Example 10.5.4

Let V be finite-dimensional with orthonormal basis $\mathfrak{B} = \{v_1, \ldots, v_n\}$ (Gram–Schmidt), and let $\alpha_{\mathfrak{B}} : V \to F^n$ be the coordinate isomorphism, $\alpha_{\mathfrak{B}}(v_i) = e_i$. Then $\langle \alpha_{\mathfrak{B}}(v_i), \alpha_{\mathfrak{B}}(v_j)\rangle = \delta_{ij} = \langle v_i, v_j\rangle$, and in general, for $v = \sum a_i v_i$, $w = \sum b_i v_i$, Observation 10.4.1(b) gives $\langle \alpha_{\mathfrak{B}}(v), \alpha_{\mathfrak{B}}(w)\rangle = \sum a_i \bar{b}_i = \langle v, w\rangle$. So $\alpha_{\mathfrak{B}}$ is an inner product space isomorphism onto F^n with the dot product.

Theorem 10.5.5 (Classification of finite-dimensional inner product spaces)

For finite-dimensional inner product spaces V, W over F the following are equivalent: (1) V and W are isomorphic as inner product spaces; (2) $\dim V = \dim W$.

Proof. (1)⇒(2): an isomorphism of vector spaces preserves dimension. (2)⇒(1): if $\dim V = \dim W = n$, both are isomorphic to F^n with the dot product (Example 10.5.4). Explicitly, for orthonormal bases $\mathfrak{B}, \mathfrak{C}$, $(\alpha_{\mathfrak{C}})^{-1} \circ \alpha_{\mathfrak{B}}$ is an inner product space isomorphism mapping $\mathfrak{B}$ to $\mathfrak{C}$. ■

Exercise 10.5.6

Write explicit inner product space isomorphisms (a) from $\mathbb{R}^2$ with the dot product to $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (b) from F^4 with the dot product to $\mathfrak{M}_{2,2}(F)$ with the trace form of Exercise 10.1.14.

Solution. (a) Send e_1, e_2 to the orthonormal basis of Exercise 10.2.13(c): $\varphi(a,b) = \frac{a}{\sqrt{2}}(1,0) + \frac{b}{\sqrt{6}}(1,2)$. (b) The matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ are orthonormal for $\operatorname{tr}(A^t\bar{B})$; send $e_1, \dots, e_4$ to them, i.e. $(a,b,c,d)^t \mapsto \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. □

Now return to rigid motions (Exercise 10.2.19). For $F = \mathbb{R}$ the results of §9.2 carry over:

Theorem 10.5.7

Let V be a finite-dimensional inner product space over $\mathbb{R}$. (a) A rigid motion L of V with $L(0) = 0$ is linear. (b) Every rigid motion of V is a translation composed with a linear rigid motion. (c) Every rigid motion of V is a bijection.

First proof. V is isomorphic as an inner product space to $\mathbb{R}^n$ with the dot product (Theorem 10.5.5), and the statements are Theorem 9.2.11 and Corollary 9.2.12. ■

Second proof. The second proof of Theorem 9.2.11(a) works verbatim with an orthonormal basis in place of $\mathcal{E}$; (b), (c) follow as before. ■

Exercise 10.5.8

Let $\gamma : \mathbb{C}^n \to \mathbb{R}^{2n}$, $\gamma((a_1 + ib_1, \dots, a_n + ib_n)^t) = (a_1, b_1, \dots, a_n, b_n)^t$, with the dot products on both spaces. (a) γ is an isomorphism of $\mathbb{R}$-vector spaces.
(b) $\|\gamma(X) - \gamma(Y)\| = \|X - Y\|$.
(c) If M is a rigid motion of $\mathbb{C}^n$, then $\gamma \circ M \circ \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$; hence M is a bijection.

Solution. (a) Clear.

(b) $|a + ib|^2 = a^2 + b^2$ coordinatewise.

(c) Distances are preserved by γ, M, γ^{-1}, so $\gamma M \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$, a bijection by Corollary 9.2.12; then $M = \gamma^{-1}(\gamma M \gamma^{-1})\gamma$ is a bijection. □

10.6 Orthogonal groups and unitary groups

Motivation. The orthogonal operators of Chapter 9 generalize to $\mathbf{O}(V)$ (real) and $\mathbf{U}(V)$ (complex): linear operators preserving distance, length, inner product and orthonormal bases. Their matrices form $\mathbf{O}(n)$ and $\mathbf{U}(n)$, characterised by $A^*A = I$.

Throughout this section spaces are finite-dimensional.

Definition 10.6.1 (Orthogonal and unitary groups)

(a) For a real inner product space V: $\mathbf{O}(V) = \mathbf{O}(V, \langle\cdot,\cdot\rangle) = \{L \in \mathfrak{L}(V,V) : \|Lv - Lw\| = \|v - w\| \ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**.
(b) For a complex inner product space V: $\mathbf{U}(V) = \mathbf{U}(V, \langle\cdot,\cdot\rangle)$ is defined the same way, the **unitary group**; its elements are **unitary operators**.

Different inner products on V give different groups. Elements of these groups are injective linear maps, hence invertible.

Exercise 10.6.2

For $G = \mathbf{O}(V)$ or $\mathbf{U}(V)$: (a) $I_V \in G$;
(b) $L, M \in G \Rightarrow L \circ M \in G$;
(c) $L \in G \Rightarrow L^{-1} \in G$.

Solution. As in Exercise 9.2.7. □

Definition 10.6.3 (Unitary matrices)

With the Hermitian dot product on $\mathbb{C}^n$, $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : L_A \in \mathbf{U}(\mathbb{C}^n)\} = \{[L]_{\mathcal{E}}^{\mathcal{E}} : L \in \mathbf{U}(\mathbb{C}^n)\}$, the (complex) **unitary group**; its elements are **unitary matrices**. Likewise $\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : L_A \in \mathbf{O}(\mathbb{R}^n)\}$.

Remark 10.6.4

Matrix versions of $\mathbf{O}(V)$, $\mathbf{U}(V)$ give nothing new: if $\mathfrak{B}$ is an orthonormal basis of n-dimensional V, then $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{U}(V)\} = \mathbf{U}(n)$ (similarly for $\mathbf{O}$). This is essentially Exercise 10.6.11 (compare Exercise 9.6.11); see also §13.3.

Observation 10.6.5

For an n-dimensional inner product space V and $L \in \mathfrak{L}(V,V)$ the following are equivalent: (1) $\|Lv - Lw\| = \|v - w\|$ (i.e. $L \in \mathbf{O}(V)$ or $\mathbf{U}(V)$); (2) $\|Lv\| = \|v\|$; (3) $\langle Lv, Lv\rangle = \langle v, v\rangle$; (4) $\langle Lv, Lw\rangle = \langle v, w\rangle$; (5) if $\mathfrak{B}$ is an orthonormal basis, so is $L(\mathfrak{B})$.

Proof. As in Observation 9.3.4 (for (5)⇒(2) use Exercise 10.4.4), except (3)⇒(4) when $F = \mathbb{C}$. Expanding $\langle L(v+w), L(v+w)\rangle = \langle v+w, v+w\rangle$ gives $\langle Lv, Lw\rangle + \langle Lw, Lv\rangle = \langle v, w\rangle + \langle w, v\rangle$, i.e. $2\,\mathrm{Re}\langle Lv, Lw\rangle = 2\,\mathrm{Re}\langle v, w\rangle$. Expanding $\langle L(v+iw), L(v+iw)\rangle = \langle v+iw, v+iw\rangle$ gives $-i\langle Lv, Lw\rangle + i\langle Lw, Lv\rangle = -i\langle v, w\rangle + i\langle w, v\rangle$, i.e. $2\,\mathrm{Im}\langle Lv, Lw\rangle = 2\,\mathrm{Im}\langle v, w\rangle$. ■

Exercise 10.6.6

One may add to Observation 10.6.5: (2)′ $\|v\| = 1 \Rightarrow \|Lv\| = 1$; (5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. As in Exercise 9.3.5. □

Definition 10.6.7 (Adjoint matrix)

For $A \in \mathfrak{M}_{m,n}(F)$, $A^* = \overline{A^t} = \bar{A}^t$ is the **adjoint matrix** (for real A, $A^* = A^t$). (It is unrelated to the classical adjoint of §6.7.)

Exercise 10.6.8

For $A, B \in \mathfrak{M}_{m,n}(F)$, $C \in \mathfrak{M}_{n,r}(F)$, $c \in F$: (a) $(A+B)^* = A^* + B^*$;
(b) $(cA)^* = \bar{c}A^*$;
(c) $(A^*)^* = A$;
(d) $(AC)^* = C^*A^*$.

Solution. Conjugation and transposition are additive, conjugation is multiplicative, $(AC)^t = C^tA^t$, and both operations are involutions. □

Exercise 10.6.9

For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ (Hermitian dot product on $\mathbb{C}^n$) the following are equivalent: (1) $\|AX - AY\| = \|X - Y\|$ (i.e. $A \in \mathbf{U}(n)$); (2) $\|AX\| = \|X\|$; (3) $\langle AX, AX\rangle = \langle X, X\rangle$; (4) $\langle AX, AY\rangle = \langle X, Y\rangle$; (5) the columns form an orthonormal basis of $\mathbb{C}^n$; (6) $A^t\bar{A} = I = \bar{A}A^t$, i.e. $A^*A = I = AA^*$, $A^{-1} = A^*$; (7) $A^t \in \mathbf{U}(n)$; (7)′ $\bar{A} \in \mathbf{U}(n)$; (7)″ $A^* \in \mathbf{U}(n)$; (8) the rows form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{C})$.

Solution. (1)–(5) is Observation 10.6.5 with $\mathfrak{B} = \mathcal{E}$. The (i,j) entry of A^*A is $\overline{\langle [A]^i, [A]^j\rangle}$, which gives (5)⇔(6). (6) is preserved by $A \mapsto A^t, \bar{A}, A^*$ (take transposes and conjugates of $A^*A = I = AA^*$), giving (7), (7)′, (7)″. (8) is (5) for A^t. □

Exercise 10.6.10

For $G = \mathbf{O}(n)$ or $\mathbf{U}(n)$: (a) $I_n \in G$;
(b) $A, B \in G \Rightarrow AB \in G$;
(c) $A \in G \Rightarrow A^{-1} \in G$.

Solution. $(AB)^*AB = B^*A^*AB = I$; $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I$. □

So $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : A^*A = I\}$. One would like to write $\mathbf{U}(V) = \{L : L^* \circ L = I\}$, but how should L^* be defined? (See Question 10.7.2.)

Exercise 10.6.11

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases of an n-dimensional complex inner product space V, then $[I]_{\mathfrak{C}}^{\mathfrak{B}} \in \mathbf{U}(n)$.

Solution. The j-th column of $[I]_{\mathfrak{C}}^{\mathfrak{B}}$ is $[w_j]_{\mathfrak{B}}$ for $w_j \in \mathfrak{C}$. By Observation 10.4.1(b), $\langle [w_i]_{\mathfrak{B}}, [w_j]_{\mathfrak{B}}\rangle = \langle w_i, w_j\rangle = \delta_{ij}$ (dot product of coordinates), so the columns are orthonormal. □

Exercise 10.6.12

If $A \in \mathbf{U}(n)$, then $|\det A| = 1$.

Solution. $1 = \det(A^*A) = \overline{\det A} \det A = |\det A|^2$. □

Definition 10.6.13 (Special unitary group)

$\mathbf{SU}(n) = \{A \in \mathbf{U}(n) : \det A = 1\}$.

Exercise 10.6.14

(a) $\mathbf{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$. What is $\mathbf{U}(2)$?

(b) Are the elements of $\mathbf{SU}(2)$ diagonalizable over $\mathbb{C}$? Those of $\mathbf{U}(2)$?

Solution. (a) Let $A = \left(\begin{smallmatrix} \alpha & z \\ \beta & w \end{smallmatrix}\right) \in \mathbf{SU}(2)$. The first column is a unit vector, $|\alpha|^2+|\beta|^2 = 1$. The second column $(z, w)^t$ is a unit vector orthogonal to it; the orthogonal complement of $(\alpha, \beta)^t$ is spanned by $(-\bar{\beta}, \bar{\alpha})^t$, so $(z, w) = \lambda(-\bar{\beta}, \bar{\alpha})$ with $|\lambda| = 1$, and $\det A = \lambda(|\alpha|^2 + |\beta|^2) = \lambda = 1$. Conversely such matrices are unitary with determinant 1. Hence $\mathbf{U}(2) = \left\{ \left(\begin{smallmatrix} \alpha & -\lambda\bar{\beta} \\ \beta & \lambda\bar{\alpha} \end{smallmatrix}\right) : |\alpha|^2 + |\beta|^2 = 1,\ |\lambda| = 1 \right\}$.

(b) Yes in both cases: unitary matrices are normal and hence unitarily diagonalizable (Chapter 15). Directly: a unitary A has a unit eigenvector v_1; complete it to an orthonormal basis $\{v_1, v_2\}$. Then $Av_2 \perp Av_1 = \lambda v_1$, so $Av_2 \in \langle v_2 \rangle$, and $\{v_1, v_2\}$ is an eigenbasis. □

For $n \geq 3$ solving the quadratic equations is hopeless; we will need the statement "an inner product can be represented by a matrix", taken up in Chapter 13.

10.7 The adjoint matrix and its applications

Motivation. The identity $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ is the heart of the adjoint. It yields $F^m = \operatorname{im} A \oplus \ker A^*$, the normal equations $A^*AX = A^*B$ of least squares, and minimal solutions $X_0 = A^*Y_0$.

Observation 10.7.1

With dot products on F^n, F^m and $A \in \mathfrak{M}_{m,n}(F)$: $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ and $\langle Y, AX \rangle = \langle A^*Y, X \rangle$ for $X \in F^n$, $Y \in F^m$.

Proof. $\langle AX, Y \rangle = (AX)^t\bar{Y} = X^tA^t\bar{Y}$, and since $\overline{A^*} = A^t$, $\langle X, A^*Y \rangle = X^t\overline{A^*Y} = X^tA^t\bar{Y}$. The second identity is the conjugate of the first. ■

Question 10.7.2

For finite-dimensional inner product spaces V, W and $L \in \mathfrak{L}(V, W)$, does there exist $L^* \in \mathfrak{L}(W, V)$ with $\langle Lv, w \rangle = \langle v, L^*w \rangle$ for all $v \in V$, $w \in W$? What does it mean?

For the rest of this section F^n carries the dot product.

Least squares approximation (LSA). If $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) has no solution, we seek X_0 minimizing $\|AX_0 - B\|$, an **LSA solution**. Example: given distinct $a_0, \ldots, a_m$ and arbitrary $b_0, \ldots, b_m$, find $g(t) = c_nt^n + \cdots + c_0 \in \mathcal{P}_n(F)$ $(m > n)$ minimizing $\sum_i |g(a_i) - b_i|^2$. With the Vandermonde-type matrix $A = (a_i^j)$, $X = (c_0, \ldots, c_n)^t$, $B = (b_0, \ldots, b_m)^t$, this is $\|AX - B\|^2$; for $m = n$ it is Lagrange interpolation.

Proposition 10.7.3

For $A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$: (a) an LSA solution X_0 ($\|AX_0 - B\| \leq \|AX - B\|$ for all X) always exists; (b) if L_A is injective ($\operatorname{rk} A = n$), it is unique.

Proof. By the closest vector problem (Observation 10.4.2), $F^m = \operatorname{im} A \oplus (\operatorname{im} A)^\perp$, and $B = AX_0 + C$ with $C \in (\operatorname{im} A)^\perp$ gives the unique vector AX_0 of $\operatorname{im} A$ closest to B. If L_A is injective, X_0 is determined by AX_0. ■

Proposition 10.7.4

The solutions of $(A^*A)X = A^*B$ are exactly the LSA solutions.

Proof. X_0 is an LSA solution iff $AX_0 - B \in (\operatorname{im} A)^\perp$, i.e. $0 = \langle AX, AX_0 - B\rangle = \langle X, A^*(AX_0 - B)\rangle$ for all X (Observation 10.7.1), i.e. $A^*(AX_0 - B) = 0$. ■

Exercise 10.7.5

Find the least squares line for the points $(1,1)$, $(2,3)$, $(3,3)$, $(4,2)$ of $\mathbb{R}^2$.

Solution. For $y = ct + d$: $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$, $B = (1,3,3,2)^t$, $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*B = (24,9)^t$. Solving, $(c,d) = (\frac{3}{10}, \frac{3}{2})$: $y = 0.3t + 1.5$. □

Exercise 10.7.6

Solve the LSA problem for $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 6$.

Solution. The coefficient matrix has rank 2, so the normal equations $(A^*A)X = A^*B$ have a line of solutions: $X = (\frac{47}{42}, \frac{4}{7}, 0)^t + t(5,-4,7)^t$, $t \in \mathbb{R}$. □

Observation 10.7.7

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk}(A^*A) = \operatorname{rk} A$;
(b) if $m \geq n$ and $\operatorname{rk} A = n$, then $A^*A \in \mathfrak{M}_{n,n}(F)$ is invertible.

Proof. (a) By the dimension theorem it suffices that $\ker(A^*A) = \ker A$. If $A^*AX = 0$, then $0 = \langle X, A^*AX\rangle = \langle (A^*)^*X, AX\rangle = \langle AX, AX\rangle$ (Observation 10.7.1, Exercise 10.6.8(c)), so $AX = 0$. (b) follows. ■

Exercise 10.7.8

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk} A = \operatorname{rk} A^t = \operatorname{rk} \bar{A} = \operatorname{rk} A^*$;
(b) $\operatorname{rk}(AA^*) = \operatorname{rk} A$.

Solution. (a) Rank theorem and Exercise 10.3.10.

(b) Observation 10.7.7(a) for A^* and (a). □

Observation 10.7.9

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{im}(A^*) = (\ker A)^\perp$, i.e. $\ker A = (\operatorname{im} A^*)^\perp$;
(b) $\operatorname{im} A = (\ker A^*)^\perp$, i.e. $\ker A^* = (\operatorname{im} A)^\perp$.

Proof. The four equalities are equivalent by Exercise 10.3.6(a) and by replacing A with A^*. We show $\ker A = (\operatorname{im} A^*)^\perp$. If $AX = 0$, then $\langle X, A^*Y\rangle = \langle AX, Y\rangle = 0$. If $X \in (\operatorname{im} A^*)^\perp$, then $\langle Y, AX\rangle = \langle A^*Y, X\rangle = 0$ for all Y, so $AX = 0$. ■

Minimal solution problem. If $AX = B$ is solvable, find the solution of smallest norm (the foot of the perpendicular from 0 to the solution set).

Proposition 10.7.10

Let $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) have a solution. (a) There is a unique minimal solution X_0 ($AX_0 = B$ and $\|X_0\| \le \|X\|$ for every solution X).
(b) X_0 is the unique solution lying in $\mathrm{im}(A^*)$.
(c) If $(AA^*)Y_0 = B$, then $X_0 = A^*Y_0$.

Proof. (i) Let $AX_1 = B$. By Observation 10.7.9, $F^n = \mathrm{im}(A^*) \oplus \ker A$, so $X_1 = A^*Y_0 + Z$ with $Z \in \ker A$. Put $X_0 = A^*Y_0$; then $AX_0 = AX_1 = B$, and X_0 is a solution in $\mathrm{im}\, A^*$.

(ii) Every solution is $X = X_0 + Z'$ with $Z' \in \ker A = (\mathrm{im}\, A^*)^\perp$, so $X_0 \perp Z'$ and $\|X\|^2 = \|X_0\|^2 + \|Z'\|^2 \ge \|X_0\|^2$, with equality iff $Z' = 0$.

(iii) If $X \in \mathrm{im}\, A^*$ is a solution, then $X_0 - X \in \mathrm{im}\, A^* \cap \ker A = 0$. (c) follows: $A^*Y_0 \in \mathrm{im}\, A^*$ is a solution. ■

So: solve $(AA^*)Y = B$ for any Y_0; then $X_0 = A^*Y_0$.

Exercise 10.7.11

Find the minimal solutions of (a) $x + 3y + z = 3$, $2x - y - 2z = 2$;
(b) $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 7$.

Solution. (a) $AA^* = \left(\begin{smallmatrix} 11 & -3 \\ -3 & 9 \end{smallmatrix}\right)$, and $Y_0 = (AA^*)^{-1}(3,2)^t$ gives $X_0 = A^*Y_0 = (\frac{19}{18}, \frac{34}{45}, -\frac{29}{90})^t$. (b) The third equation is the first plus twice the second, so the system is consistent with the same solution set as (a), and the minimal solution is the same. □